



SMART ENERGY THEFT DETECTION SYSTEM USING IOT AND AI



A PROJECT REPORT

Submitted by

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*in partial fulfillment of the requirements for the award of
the course*

AMB1321 – INTERNET OF THINGS

**DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE
AND MACHINE LEARNING)**

**K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)**

SAMAYAPURAM- 621112

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K. RAMAKRISHNAN COLLEGE OF TECHNOLOGY**(AUTONOMOUS)****SAMAYAPURAM- 621112****BONAFIDE CERTIFICATE**

The work embodied in the present project report entitled “**SMART ENERGY THEFT DETECTION SYSTEM USING IOT AND AI**” has been carried out by the students **SAMEEHA J, SUDHARSON S R, VETHAABINAYA S, VIMAL AANANTH B R**. The work reported here in is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

In modern power distribution systems, electricity theft has become a major issue that leads to financial losses and affects the efficiency of energy supply. Traditional monitoring methods are not capable of detecting abnormal energy usage in real time, making it difficult to identify unauthorized consumption. To address this problem, advanced technologies such as IoT and AI are used to develop smart monitoring systems. This project, titled “Smart Energy Theft Detection System Using IoT and AI”, focuses on monitoring and analyzing real-time energy consumption. The system uses an ACS712 current sensor to measure current flow and an ESP32 microcontroller to process the data. The ESP32 continuously reads the sensor values and calculates energy usage, enabling real-time monitoring. The system applies statistical threshold analysis to define normal energy consumption levels and identify deviations. Anomaly detection techniques are used to detect unusual patterns such as sudden drops or spikes in current. This hybrid approach improves detection accuracy and reduces false alarms. Whenever abnormal behavior is detected, the system generates alerts using an LCD display and a buzzer. The LCD shows the current status, while the buzzer provides an immediate warning. The IoT capability of the ESP32 also allows data transmission for monitoring and analysis. The proposed system is cost-effective and suitable for real-time applications. It provides a reliable solution for detecting electricity theft and improving energy management.

Keywords : Electricity Theft Detection, Internet of Things (IoT), Artificial Intelligence (AI), ESP32 Microcontroller, ACS712 Current Sensor, Real-Time Monitoring, Anomaly Detection, Smart Energy System, Energy Management, Power Distribution Systems.

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LIST OF ABBREVIATIONS

AI	–	Artificial Intelligence
IoT	–	Internet of Things
ESP32	–	Electronic System Processor 32
ACS712	–	Current Sensor Module
LCD	–	Liquid Crystal Display
ML	–	Machine Learning
ADC	–	Analog to Digital Converter
AC	–	Alternating Current
DC	–	Direct Current
Wi-Fi	–	Wireless Fidelity
CPU	–	Central Processing Unit
VCC	–	Voltage Common Collector
GND	–	Ground
PWM	–	Pulse Width Modulation
kWh	–	Kilowatt Hour

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The Electricity theft is a serious problem in modern power distribution systems, causing significant financial losses and affecting the stability and reliability of energy supply. Traditional energy meters are designed to measure only total consumption and do not have the capability to detect abnormal or unauthorized usage in real time. Manual inspection methods are time-consuming, costly, and often inefficient, especially in large-scale power networks. These limitations make it difficult to monitor energy usage effectively and prevent electricity theft.

With the advancement of IoT and AI, smart energy monitoring systems have emerged as a promising solution. IoT enables continuous monitoring of energy consumption through sensors, while AI techniques analyze usage patterns to identify anomalies. These technologies provide an automated and intelligent approach for detecting unusual energy behavior, improving efficiency, and reducing losses in power distribution systems. This project is designed to monitor real-time current consumption using an ACS712 current sensor and process the data using an ESP32 microcontroller.

The system continuously collects data and applies statistical threshold analysis along with anomaly detection techniques to identify abnormal patterns such as sudden drops or irregular consumption. When suspicious activity is detected, alerts are generated using an LCD display and buzzer, enabling immediate action. This system provides a cost-effective, efficient, and scalable solution for electricity theft detection. It improves energy management, supports real-time monitoring, and can be extended for smart grid applications and advanced power distribution systems.

1.2 OBJECTIVE

The main objective of this project is to design and develop a smart energy theft detection system using IoT and Artificial Intelligence that can effectively monitor real-time energy consumption and identify abnormal usage patterns. The system aims to provide a reliable and automated solution for detecting electricity theft by continuously analyzing energy data collected from sensors.

A key focus of this project is the implementation of real-time monitoring using an ACS712 current sensor and ESP32 microcontroller, which enables continuous data acquisition and processing. The system applies statistical threshold analysis, including mean and standard deviation, to determine the normal range of energy consumption and detect deviations from expected values. In addition, anomaly detection techniques are used to identify unusual patterns such as sudden drops or spikes in energy usage, improving detection accuracy.

This project is to develop a smart energy theft detection system using IoT and AI that can monitor real-time energy consumption and detect abnormal usage patterns. The system aims to analyze data using statistical and anomaly detection techniques and provide instant alerts for unauthorized energy usage.

The system also aims to provide instant alert mechanisms using an LCD display and buzzer, allowing quick identification of suspicious activity. Furthermore, the integration of IoT enables efficient data handling and monitoring, making the system suitable for real-time applications.

Overall, the project focuses on developing a cost-effective, scalable, and efficient solution that can be used in modern energy management systems and extended for smart grid applications.

CHAPTER 2

LITERATURE SURVEY

2.1 AI-BASED ENERGY THEFT DETECTION USING IOT AND ML

M. K. A. Kumar, R. Sharma, P. Singh

Published Year: 2025

This study presents an IoT-based system for detecting electricity theft using machine learning techniques. The system collects real-time energy consumption data through smart meters and analyzes it using classification algorithms to identify abnormal usage patterns. Machine learning models are trained to differentiate between normal and suspicious consumption behavior. The system enables automated monitoring and provides alerts when irregularities are detected. This approach highlights the importance of integrating IoT and AI for improving efficiency in power distribution systems.

Merits

- Enables real-time monitoring of energy consumption
- Uses machine learning for accurate anomaly detection
- Provides automated alerts for suspicious activity
- Improves efficiency of power management systems

Demerits

- Requires large datasets for better model training
- Performance may be affected by noisy data
- Initial setup cost may be high.

2.2 SMART ENERGY MONITORING AND THEFT DETECTION USING IOT

S. Reddy, K. Verma, L. Gupta

Published Year: 2026

This research focuses on an IoT-based smart energy monitoring system that detects electricity theft using real-time data analysis. Sensors are used to measure current and transmit data to a microcontroller, which processes the information and identifies abnormal patterns. The system uses threshold-based detection to determine unusual energy consumption and generates alerts when theft is suspected. The study demonstrates a low-cost and efficient solution for real-time monitoring and theft detection.

Merits

- Cost-effective solution for energy monitoring
- Real-time data collection and analysis
- Simple implementation using IoT devices
- Suitable for small-scale and smart grid applications

Demerits

- Accuracy Limited accuracy without advanced AI techniques
- Threshold-based methods may generate false alerts
- Requires proper calibration for accurate results

2.3 MACHINE LEARNING BASED ANOMALY DETECTION IN POWER SYSTEMS

J. Lee, H. Kim, S. Park

Published Year: 2025

This study presents a machine learning-based anomaly detection system for identifying irregular energy consumption patterns in power systems. The system uses statistical features and supervised learning algorithms to classify energy usage as normal or abnormal. The model is trained using historical data and can detect sudden variations in energy consumption. The approach improves detection accuracy and reduces manual monitoring efforts.

Merits

- Detects High accuracy in detecting abnormal patterns
- Reduces need for manual inspection
- Efficient analysis of large datasets
- Supports automated monitoring systems

Demerits

- Requires training with large datasets
- Performance depends on data quality
- Complex model implementation

2.4 IOT-BASED REAL-TIME ENERGY MONITORING SYSTEM

M. Patel, D. Shah, N. Joshi

Published Year: 2026

This research introduces an IoT-based real-time energy monitoring system that uses sensors and microcontrollers to track electricity usage. The system collects data continuously and displays consumption values for monitoring purposes. It provides a basic framework for identifying abnormal usage by comparing real-time data with expected values. The system is simple, efficient, and suitable for practical applications.

Merits

- Continuous monitoring of energy consumption
- Easy implementation and low cost
- Provides real-time data visualization
- Useful for basic monitoring applications.

Demerits

- Limited detection capability without AI integration
- Accuracy depends on sensor performance
- Cannot detect complex usage patterns

2.5 HYBRID AI-BASED ENERGY THEFT DETECTION SYSTEM

R. Gupta, A. Mehta, S. Jain

Published Year: 2025

This study proposes a hybrid energy theft detection system that combines statistical methods and machine learning techniques. The system first applies threshold analysis to detect initial anomalies and then uses machine learning models to confirm suspicious behavior. This two-stage approach improves detection accuracy and reduces false positives. The system demonstrates the effectiveness of combining multiple techniques for reliable energy monitoring.

Merits

- Improved accuracy using hybrid approach
- Reduces false alarms through verification
- Combines statistical and AI techniques
- Suitable for real-time applications

Demerits

- Increased complexity in implementation
- Requires proper tuning of parameters
- Depends on availability of training data

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Traditional methods for monitoring electricity consumption mainly rely on conventional energy meters that record total usage without providing real-time analysis. These systems are not capable of detecting abnormal or unauthorized energy usage patterns. Electricity theft is usually identified through manual inspections or periodic checks, which are time-consuming, inefficient, and prone to human error. In some advanced systems, basic monitoring tools are used to observe energy consumption; however, they lack intelligent analysis capabilities. These systems cannot effectively identify sudden variations or irregular patterns in energy usage. Moreover, traditional approaches do not support real-time alerts, making it difficult to take immediate action when theft occurs.

Due to the absence of automation and intelligent processing, existing systems are not suitable for modern power distribution networks. They fail to provide accurate, efficient, and scalable solutions for detecting electricity theft.

3.1.1 Demerits

- No real-time monitoring of energy consumption
- Unable to detect abnormal usage patterns effectively
- Depends on manual inspection methods
- Time-consuming and inefficient
- Lack of automated alert mechanisms
- Not suitable for large-scale power systems

3.2 PROPOSED SYSTEM

The proposed system presents a smart energy theft detection solution using IoT and Artificial Intelligence. It is designed to monitor real-time energy consumption and identify abnormal usage patterns automatically. The system uses an ACS712 current sensor to measure current flow and an ESP32 microcontroller to process the data. The ESP32 continuously collects current values and analyzes the data using statistical threshold analysis and anomaly detection techniques. The statistical method determines the normal consumption range using mean and standard deviation, while anomaly detection identifies unusual variations such as sudden drops or spikes in energy usage. This hybrid approach improves detection accuracy and reduces false alarms. The system generates alerts using an LCD display and buzzer when abnormal activity is detected. The LCD displays the current status, while the buzzer provides an immediate warning. This enables quick identification and response to suspicious energy usage. The proposed system is cost-effective, efficient, and suitable for real-time applications. It provides a reliable solution for electricity theft detection and can be extended for smart grid systems and advanced energy management.

3.2.1 Merits

- Provides real-time monitoring of energy consumption
- Detects abnormal usage patterns effectively
- Generates instant alerts using LCD and buzzer
- Reduces manual effort and human error
- Cost-effective and easy to implement
- Suitable for smart grid and IoT applications

CHAPTER 4

SYSTEM SPECIFICATION

4.1 HARDWARE SPECIFICATIONS

- Microcontroller : ESP32
- Sensor : ACS712 Current Sensor
- Display : 16x2 LCD Display
- Alert Device : Buzzer
- Switching Device : Relay Module
- Input : Electrical Load (Bulb)
- Power Supply : 5V DC Supply
- Development Setup : Breadboard and Jumper Wires

4.2 SOFTWARE SPECIFICATIONS

- Programming Language : Embedded C / C++
- Development Environment : Arduino IDE
- Microcontroller Platform : ESP32 Board Package
- Libraries Used : LiquidCrystal I2C, Wi-Fi Library
- Operating System : Windows

CHAPTER 5

SYSTEM DESIGN

5.1 SYSTEM ARCHITECTURE

The system architecture represents the overall structure of the smart energy theft detection system using IoT and Artificial Intelligence. The system is designed to monitor, process, and analyze energy consumption in real time and detect abnormal usage effectively. It consists of multiple stages including data acquisition, processing, detection, alert generation, and control.

The system begins with the data acquisition stage, where the ACS712 current sensor is used to measure real-time current consumption from the electrical load. The sensor continuously captures current values and converts them into corresponding analog signals. These signals are then sent to the ESP32 microcontroller for further processing.

The ESP32 acts as the central processing unit of the system. It reads the sensor data through its analog input pins and converts it into digital values. The microcontroller processes the data to calculate energy consumption and monitors variations in current values. This enables continuous tracking of energy usage patterns in real time.

The collected data is analyzed using statistical threshold methods and anomaly detection techniques. The statistical method calculates the mean and standard deviation to determine the normal range of energy consumption. Any deviation beyond the defined threshold is considered suspicious. In addition, anomaly detection techniques are used to identify irregular patterns such as sudden drops or unexpected spikes in current values. This hybrid approach improves detection accuracy and reduces false alarms.

System Architecture - Smart Energy Theft Detection Using IoT and AI

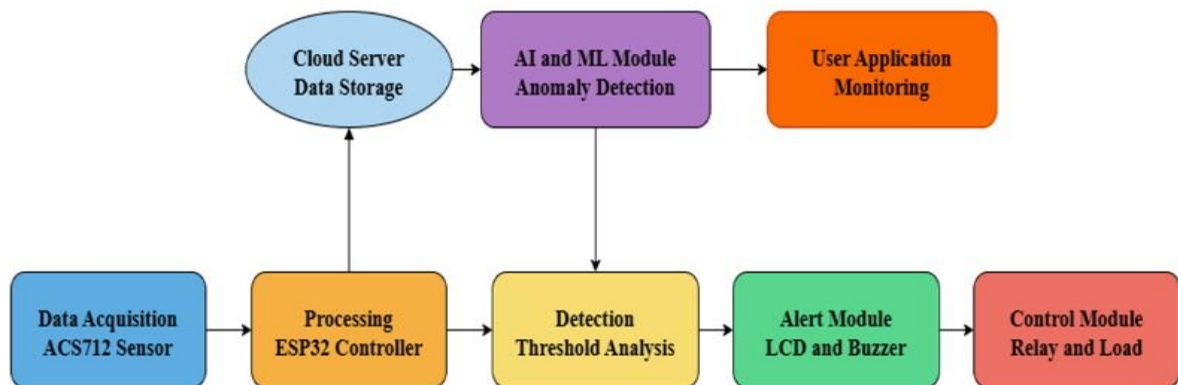


Figure.5.1 System Architecture

The alert module consists of a 16x2 LCD display and a buzzer. The LCD displays the real-time status of energy consumption, including whether the system is operating under normal or abnormal conditions. When an abnormal condition is detected, the buzzer is activated to provide an immediate alert. This helps in quickly identifying electricity theft or suspicious activity.

Finally, the control module consists of a relay connected to the load. The relay can be used to control the electrical supply if required. The overall system provides a structured and efficient approach for real-time monitoring, detection, and alert generation.

CHAPTER 6

MODULE DESCRIPTION

6.1 METHODOLOGY

6.1.1 ALGORITHM IMPLEMENTATION : HYBRID STATISTICAL AND ANOMALY DETECTION METHOD

The proposed system follows a hybrid methodology that combines statistical analysis and anomaly detection techniques to identify abnormal energy usage. The system starts by collecting real-time current data using the ACS712 sensor connected to the ESP32 microcontroller. The sensor measures the current flowing through the load and sends the data to the ESP32 for processing.

In the first stage, statistical threshold analysis is applied to determine the normal range of energy consumption. The system calculates the mean (μ) and standard deviation (σ) of the collected data and defines a threshold range using $(\mu \pm 3\sigma)$. Any value outside this range is considered abnormal.

In the second stage, anomaly detection is used to identify unusual patterns such as sudden drops or irregular spikes in current values. This helps in detecting complex cases that cannot be identified using simple threshold methods alone. The combination of statistical analysis and anomaly detection improves the accuracy of the system and reduces false alerts. The system continuously monitors the data in real time and evaluates the energy consumption pattern. When abnormal behavior is detected, the system generates alerts using an LCD display and a buzzer.

The LCD displays the current status, while the buzzer provides an immediate warning. This enables quick identification of electricity theft. Overall, the methodology provides an efficient and reliable approach for monitoring energy usage and detecting abnormal patterns in real time.

6.2 DATA ACQUISITION MODULE

The data acquisition module is responsible for collecting real-time energy consumption data from the electrical system. In this project, the ACS712 current sensor is used to measure the current flowing through the load. The sensor plays a key role in capturing accurate and continuous data required for analysis.

The ACS712 sensor is connected in series with the electrical load, such as a bulb, to measure the current passing through it. It converts the current into a corresponding analog voltage signal. This analog output is then provided to the ESP32 microcontroller for further processing. The sensor is capable of detecting both small and large variations in current, making it suitable for real-time monitoring applications.

The ESP32 continuously reads the analog values from the sensor through its input pins. These values represent the real-time current consumption of the system. The data is collected at regular intervals to ensure continuous monitoring and accurate tracking of energy usage patterns. Proper calibration of the sensor is important to obtain accurate readings.

The system ensures that the sensor output is stable and reliable under normal operating conditions. Any variation in the sensor readings directly reflects changes in energy consumption, which is essential for detecting abnormal behavior.

The collected data serves as the input for further processing and analysis in the system. Accurate data acquisition is crucial for the overall performance of the system, as it directly affects the detection of abnormal energy usage.

Overall, the data acquisition module provides a reliable and continuous flow of real-time data, forming the foundation for monitoring, analysis, and detection of electricity theft.

6.3 PROCESSING MODULE

The processing module is responsible for analyzing the data collected from the current sensor and converting it into meaningful information. In this system, the ESP32 microcontroller acts as the main processing unit. It receives the analog signals from the ACS712 sensor and processes them to determine energy consumption.

The ESP32 reads the analog input values from the sensor through its input pins and converts them into digital values using its built-in analog-to-digital converter (ADC). These digital values represent the current flowing through the electrical load. The microcontroller then processes these values to monitor variations in energy usage over time.

The processing module continuously evaluates the incoming data and prepares it for further analysis. It ensures that the data is stable and consistent by filtering out small fluctuations that may occur due to noise or environmental factors. This helps in maintaining accurate and reliable readings.

In addition, the ESP32 performs calculations to identify changes in current values and tracks the overall consumption pattern. The processed data is then passed to the detection module, where it is analyzed to identify abnormal usage.

The processing module plays an important role in handling real-time data efficiently. It ensures smooth communication between different components of the system and supports continuous monitoring of energy consumption.

Overall, this module provides the necessary data processing required for accurate analysis and helps in improving the performance of the energy theft detection system.

6.4 DETECTION MODULE

Feature The detection module is responsible for identifying abnormal energy consumption patterns based on the processed data. This module plays a key role in detecting possible electricity theft by analyzing variations in current values.

The system uses a hybrid approach that combines statistical threshold analysis and anomaly detection techniques. In the first stage, the statistical method is applied to determine the normal range of energy consumption. The system calculates the mean and standard deviation of the collected data and defines a threshold range. Any value that exceeds this range is considered abnormal.

In the second stage, anomaly detection is used to analyze the pattern of energy usage. The system identifies unusual variations such as sudden drops, irregular spikes, or inconsistent current values that do not follow normal usage behavior. This helps in detecting complex cases that cannot be identified using simple threshold methods alone.

The detection module continuously monitors the processed data in real time and compares it with the defined normal range. If any abnormal condition is identified, the system flags it as a potential case of electricity theft.

This module improves the accuracy of the system by reducing false alerts and ensuring reliable detection. The combination of statistical and anomaly-based methods enhances the system's ability to detect both simple and complex irregularities in energy consumption.

Overall, the detection module provides an efficient mechanism for identifying abnormal energy usage and plays a crucial role in the functioning of the smart energy theft detection system.

6.5 ALERT MODULE

Data Preparation The alert module is responsible for providing output indications when abnormal energy usage is detected. This module helps in notifying the user about the system status in real time and enables quick response to suspicious activity.

The system uses a 16x2 LCD display and a buzzer as the main output devices. The LCD display is used to show the current status of energy consumption. It continuously displays whether the system is operating under normal or abnormal conditions. This allows the user to monitor the system easily.

When the detection module identifies abnormal behavior, the alert module is activated. The LCD displays a warning message such as “THEFT DETECTED” or abnormal status indication. At the same time, the buzzer is turned ON to provide an audible alert. This ensures that the user is immediately notified about the suspicious activity.

The alert mechanism is designed to respond quickly to changes in energy usage. It provides both visual and audio indications, making the system more effective and user-friendly. The buzzer alert helps in situations where the user may not be constantly monitoring the display.

The module operates continuously along with the system and ensures that alerts are generated whenever abnormal conditions occur. It enhances the overall reliability of the system by providing immediate feedback.

Overall, the alert module plays an important role in notifying the user and ensuring timely action in case of electricity theft or abnormal energy consumption.

6.6 CONTROL MODULE

The control module is responsible for managing the electrical load based on the system condition. It helps in preventing further electricity theft by controlling the power supply when abnormal usage is detected.

In this system, a relay module is used to control the electrical load. The relay acts as a switch that can turn the power ON or OFF based on the signal received from the ESP32 microcontroller. Under normal conditions, the relay allows the load to operate without interruption.

When the detection module identifies abnormal energy usage, the ESP32 sends a signal to the relay module. Based on this signal, the relay can disconnect the load from the power supply, thereby preventing further unauthorized consumption. This provides an additional level of protection in the system.

The relay module is connected to the load and controlled through the output pins of the ESP32. It operates automatically based on the system status, without requiring manual intervention. This ensures efficient and reliable control of the electrical supply.

The control module works in coordination with the detection and alert modules to provide a complete solution for energy theft detection and prevention. It not only identifies abnormal usage but also takes action to minimize energy loss.

Overall, the control module enhances the effectiveness of the system by enabling automatic control of the load and improving the overall security of the energy monitoring system.

CHAPTER 7

RESULT AND DISCUSSION

7.1 COMPARATIVE STUDY

The proposed smart energy theft detection system has been successfully designed and partially implemented to monitor real-time energy consumption and identify abnormal usage patterns. The system uses an ACS712 current sensor and ESP32 microcontroller to continuously collect and process current values from the electrical load. The experimental results show that the system is capable of providing stable and reliable readings under normal operating conditions.

During normal operation, the current values measured by the sensor remain consistent and are accurately displayed on the LCD. The system continuously monitors energy usage without any interruption. The buzzer remains OFF, indicating that no abnormal condition is detected.

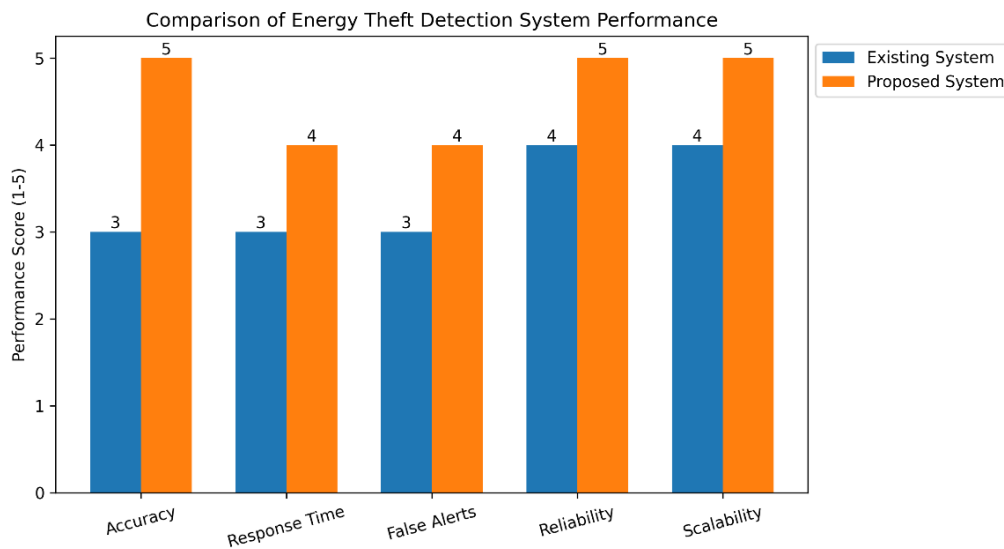


Figure.7.1 Comparative Analysis

This confirms that the system can effectively perform real-time monitoring of energy consumption.

To evaluate the detection capability, abnormal conditions such as sudden variations in current values and irregular usage patterns were considered. The system is able to identify deviations from the predefined threshold range using statistical analysis. When such variations occur, the detection module flags the condition as abnormal. As a result, the alert module is activated, displaying a warning message on the LCD and triggering the buzzer to notify the user.

The system demonstrates a fast response time in detecting changes in energy consumption. The ESP32 processes the data efficiently, and the sensor provides continuous and stable output. The combination of statistical threshold analysis and anomaly detection improves the reliability of the system and reduces the chances of false alerts.

However, the system is currently tested under controlled conditions, and advanced anomaly detection techniques are still under development. Further improvements can be made by integrating machine learning models and cloud-based monitoring for better accuracy and scalability.

Overall, the system provides a reliable and efficient solution for monitoring energy consumption and detecting abnormal usage. It demonstrates the effectiveness of combining IoT and intelligent techniques for real-time energy theft detection.

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 CONCLUSION

The proposed smart energy theft detection system presents an efficient and reliable solution for monitoring real-time energy consumption using IoT and intelligent analysis techniques. The system integrates an ACS712 current sensor and ESP32 microcontroller to continuously measure and process current values from the electrical load. This enables accurate tracking of energy usage and helps in identifying abnormal consumption patterns that may indicate electricity theft.

The system applies statistical threshold analysis to define the normal range of energy consumption based on calculated mean and standard deviation values. Any deviation from this range is considered abnormal and is further analyzed using anomaly detection techniques. This hybrid approach improves the accuracy of the system by identifying both simple and complex variations in energy usage patterns. It also reduces false alerts and enhances the reliability of detection.

The system demonstrates stable performance under normal operating conditions by providing consistent sensor readings and continuous monitoring. When abnormal behavior is detected, the system successfully generates alerts using an LCD display and buzzer, ensuring immediate notification to the user. This allows quick identification and response to suspicious activity, thereby minimizing energy loss.

In addition, the system is cost-effective, easy to implement, and suitable for real-time applications. The use of IoT technology enables efficient data handling and monitoring, making the system adaptable for future enhancements. Overall, the project successfully demonstrates the integration of IoT and intelligent techniques for detecting abnormal energy usage and improving energy management in modern power systems.

8.2 FUTURE ENHANCEMENT

The proposed smart energy theft detection system provides a reliable foundation for real-time monitoring and detection of abnormal energy usage; however, there are several areas where the system can be further enhanced to improve its performance, accuracy, and scalability. One of the major improvements that can be implemented is the integration of advanced Artificial Intelligence and Machine Learning algorithms for anomaly detection. By incorporating more sophisticated learning models, the system can better understand complex usage patterns and improve its ability to accurately identify electricity theft, even in dynamic environments.

Another important enhancement is the integration of cloud-based platforms for data storage and analysis. By transmitting real-time energy consumption data to the cloud, the system can enable remote monitoring and provide users with access to data from any location. This also allows for long-term data analysis, which can help in identifying trends and improving system performance over time. In addition, the development of mobile or web-based applications can provide users with real-time alerts, notifications, and control over the system, making it more user-friendly and accessible.

Further improvements can be made by enhancing the accuracy of the sensor readings through proper calibration and optimization techniques. This ensures that the system provides precise measurements under different operating conditions. The system can also be extended to support large-scale applications such as smart grids and industrial environments, where monitoring of multiple loads and complex energy systems is required.

Moreover, the system can be improved by incorporating secure communication protocols to ensure data protection and prevent unauthorized access. This is especially important when the system is connected to cloud platforms or remote networks. Overall, these enhancements will make the system more robust, scalable, and suitable for advanced energy management applications in the future.

APPENDIX A

SOURCE CODE

```
// ===== YOUR CREDENTIALS =====

#define BLYNK_TEMPLATE_ID "TMPL3Alo2K6lO"
#define BLYNK_TEMPLATE_NAME "Energy Meter"
#define BLYNK_AUTH_TOKEN "fPuFFqHtC1_ZiC71ddMflpqy9aefwNa4" // from
blynk.cloud
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <WiFi.h>
#include <WiFiClientSecure.h>
#include <UniversalTelegramBot.h>
#include <BlynkSimpleEsp32.h>
const char* ssid    = "iPhone";
const char* pass    = "1234567890";
#define BOT_TOKEN   "8719149611:AAFEgurcWJ9WbTJBLCkuAHXzQIrlAQIsFXg" //
from BotFather
#define CHAT_ID     "1139969911"           // from userinfobot
// =====

LiquidCrystal_I2C lcd(0x27, 16, 2);
#define CURRENT_PIN   34
#define RELAY_PIN     27
#define BUZZER_PIN    25
float voltage         = 230.0;
float sensitivity     = 0.185;
#define NOISE_FLOOR   0.08
#define LOW_CURRENT_LIMIT 0.10
#define CONFIRM_TIME  3000
bool theftTriggered = false;
```



```

bool loadWasOn    = false;
bool alertSent    = false;    // so alert sends only once
unsigned long lowStart    = 0;
unsigned long lastBlynkUpdate = 0;
int  zeroOffset    = 2048;
// Telegram client
WiFiClientSecure secured_client;
UniversalTelegramBot bot(BOT_TOKEN, secured_client);
// ----- CALIBRATE ZERO -----
int calibrateZero() {
    long sum = 0;
    for (int i = 0; i < 2000; i++) {
        sum += analogRead(CURRENT_PIN);
        delayMicroseconds(200);
    }
    return (int)(sum / 2000);
}
// ----- READ CURRENT -----
float readCurrent() {
    long sumSq = 0;
    int samples = 1000;
    for (int i = 0; i < samples; i++) {
        int raw    = analogRead(CURRENT_PIN);
        int centered = raw - zeroOffset;
        sumSq += (long)centered * centered;
        delayMicroseconds(100);
    }
    float rms    = sqrt((float)sumSq / samples);
    float voltRms = rms * (3.3 / 4095.0);
    float current = voltRms / sensitivity;
    if (current < NOISE_FLOOR) current = 0.0;

```

```

return current;
}
// ----- SEND TELEGRAM -----
void sendTelegram(String message) {
    Serial.println("Sending Telegram: " + message);
    bot.sendMessage(CHAT_ID, message, "");
}
// ----- THEFT SEQUENCE -----
void triggerTheft() {
    if (theftTriggered) return;
    theftTriggered = true;
    Serial.println("=== THEFT DETECTED ===");
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("!! THEFT !!");
    lcd.setCursor(0, 1);
    lcd.print("BYPASS DETECTED!");
    // Send Blynk notification
    Blynk.virtualWrite(V3, 1);
    Blynk.logEvent("theft_alert", " ⚡ THEFT DETECTED! Bypass active. Power will be
cut.");
    // Send Telegram message
    if (!alertSent) {
        sendTelegram(" 🚨 *THEFT DETECTED!* \n ⚡ Bypass wire active! \n 🔌 Power will be
cut in 7 seconds. \n 🟡 Smart Energy Meter Alert");
        alertSent = true;
    }
    // Buzzer beeps for 7 seconds
    unsigned long start = millis();
    while (millis() - start < 7000) {
        digitalWrite(BUZZER_PIN, HIGH); delay(300);

```

```

digitalWrite(BUZZER_PIN, LOW); delay(200);
}
digitalWrite(BUZZER_PIN, LOW);
// Cut power
digitalWrite(RELAY_PIN, LOW);
// Send power cut confirmation to Telegram
sendTelegram(" 🚫 Power CUT. Bulb turned OFF.\n Smart Energy Meter");
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("!! THEFT !!");
lcd.setCursor(0, 1);
lcd.print("Power Cut!");
Serial.println("Relay tripped. Bulb OFF.");
}
// ----- SETUP -----
void setup() {
  Serial.begin(115200);
  Wire.begin(21, 22);
  pinMode(RELAY_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  analogReadResolution(12);
  analogSetAttenuation(ADC_11db);
  digitalWrite(RELAY_PIN, HIGH);
  digitalWrite(BUZZER_PIN, LOW);
  lcd.init();
  lcd.backlight();
  // Connect WiFi first
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Connecting WiFi");
  lcd.setCursor(0, 1);
  lcd.print(ssid);

```

```

WiFi.begin(ssid, pass);
int tries = 0;
while (WiFi.status() != WL_CONNECTED && tries < 20) {
  delay(500);
  Serial.print(".");
  tries++;
}
if (WiFi.status() == WL_CONNECTED) {
  Serial.println("\nWiFi connected: " + WiFi.localIP().toString());
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("WiFi OK!");
  lcd.setCursor(0, 1);
  lcd.print(WiFi.localIP());
  delay(2000);
} else {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("WiFi FAILED");
  lcd.setCursor(0, 1);
  lcd.print("Offline mode");
  delay(2000);
}
// Telegram SSL
secured_client.setInsecure();
// Blynk connect
Blynk.config(BLYNK_AUTH_TOKEN);
Blynk.connect(3000);
// Send startup message to Telegram
sendTelegram("✅ Smart Energy Meter Online!\n📶 WiFi: " + String(ssid) + "\n🔍
Monitoring started.");

```

```

// Calibrate
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Calibrating...");
lcd.setCursor(0, 1);
lcd.print("Remove load!");
delay(2000);
zeroOffset = calibrateZero();
Serial.print("Zero offset: "); Serial.println(zeroOffset);
lcd.clear();
lcd.setCursor(0, 0);
lcd.print("Zero: "); lcd.print(zeroOffset);
lcd.setCursor(0, 1);
lcd.print("System Ready!");
delay(2000);
lcd.clear();
}
// ----- LOOP -----
void loop() {
  Blynk.run(); // keep Blynk alive
  if (theftTriggered) {
    digitalWrite(RELAY_PIN, LOW);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("!! THEFT !!");
    lcd.setCursor(0, 1);
    lcd.print("Power Cut!");
    delay(2000);
    return;
  }
  float current = readCurrent();
  float power = voltage * current;

```

```

Serial.print("Zero:"); Serial.print(zeroOffset);
Serial.print(" Current:"); Serial.print(current, 3);
Serial.print("A Power:"); Serial.println(power, 1);
// Push to Blynk every 2 seconds
if (millis() - lastBlynkUpdate > 2000) {
  Blynk.virtualWrite(V0, voltage);
  Blynk.virtualWrite(V1, current);
  Blynk.virtualWrite(V2, power);
  lastBlynkUpdate = millis();
}
// ---- NORMAL ----
if (current >= LOW_CURRENT_LIMIT) {
  lowStart = 0;
  loadWasOn = true;

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("V:"); lcd.print(voltage);
  lcd.setCursor(0, 1);
  lcd.print("I:"); lcd.print(current, 2); lcd.print("A");
  digitalWrite(RELAY_PIN, HIGH);
  digitalWrite(BUZZER_PIN, LOW);
}
// ---- LOW / ZERO ----
else {
  if (loadWasOn) {
    if (lowStart == 0) lowStart = millis();
    unsigned long lowDuration = millis() - lowStart;
    Serial.print("Low current for ");
    Serial.print(lowDuration / 1000);
    Serial.println("s");
    lcd.clear();
  }
}

```

```
lcd.setCursor(0, 0);  
  lcd.print("Low Current!");  
  lcd.setCursor(0, 1);  
  lcd.print("Check: ");  
  lcd.print(lowDuration / 1000);  
  lcd.print("s");  
  if (lowDuration >= CONFIRM_TIME) {  
    triggerTheft();  
  }  
} else {  
  lcd.clear();  
  lcd.setCursor(0, 0);  
  lcd.print("System Ready");  
  lcd.setCursor(0, 1);  
  lcd.print("No Load");  
}  
}  
delay(1000);  
}
```

APPENDIX B

SCREENSHOTS

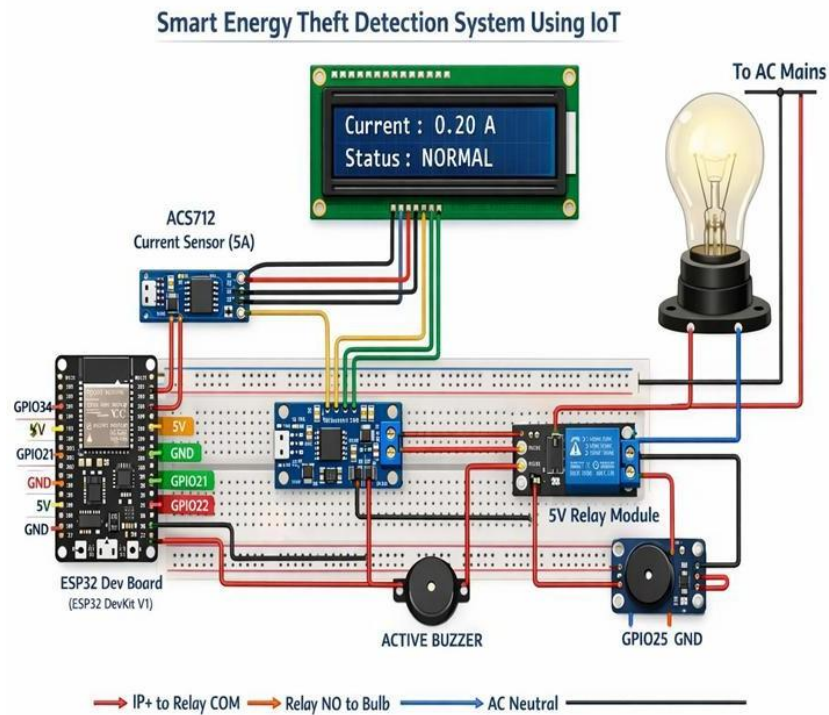


Figure.B.1 Circuit Diagram

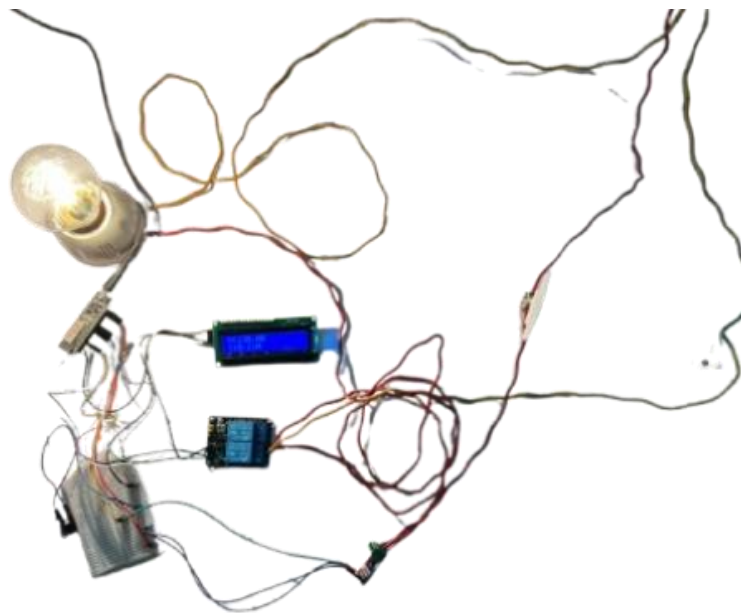


Figure.B.2 Hardware Implementation

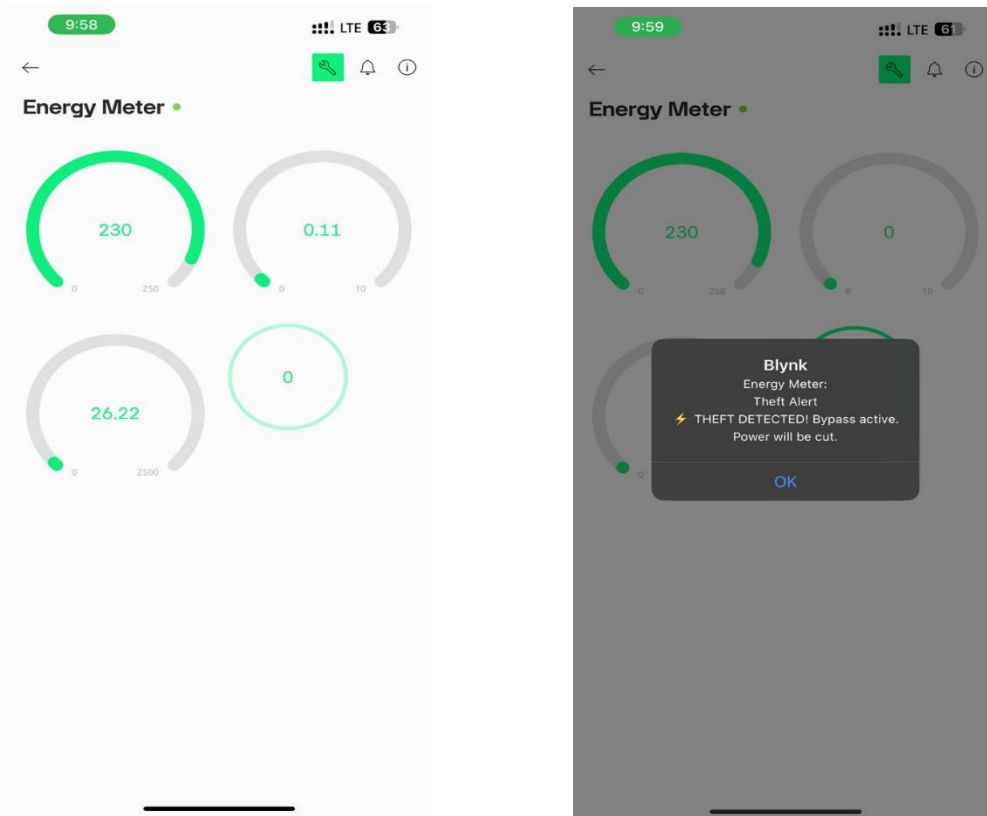


Figure.B.3 Energy Meter Diagram

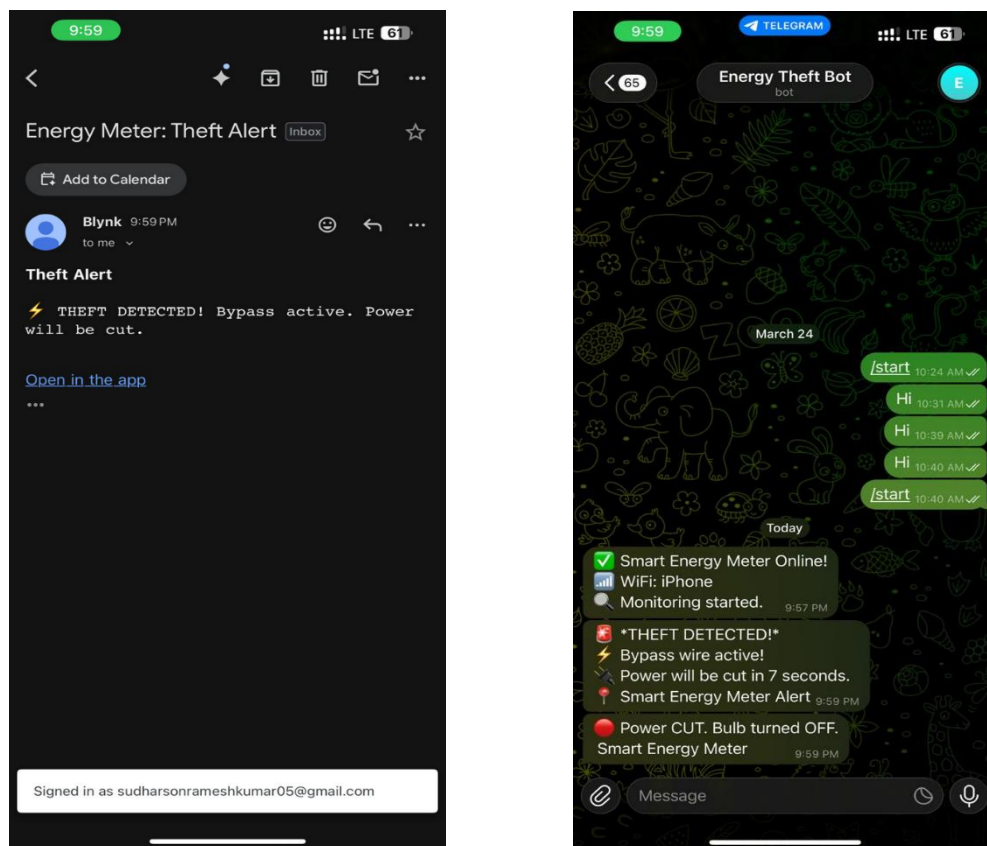


Figure.B.4 System Output / Results

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